SLR-JE-783

Seat No.	

Set P

S.Y. (B.Tech.) (Sem - II) (New) (CBCS) Examination: March/April-2024
COMPUTER SCIENCE AND ENGINEERING
Theory of Computation (BTN03402)

Day & Date: Wednesday, 22-05-2024 Max. Marks: 70
Time: 10:00 AM To 01:00 PM

Instructions: 1) Q. No. 1 is compulsory. It should be solved in the first 30 minutes in answer book Page no 03 (Starting page of the Answer Book). Each question carries one mark.
2) Don't forget to Mention question paper set (P/Q/R/S) on top of page.
3) Figures to the right indicates full marks.
4) Assume data wherever necessary.

MCQ/Objective Type Questions

Duration: 30 Minutes Marks:14

Q.1 Choose correct option **14**

1) The language of all string starting with bba and ending with abb over alphabet {a, b} is represented by the regular expression.
a) baa(a+b)*abb b) bba(a+b)*abb
c) (a+b)*abb(a+b)* d) None of the above

2) Let $\Sigma = \{a, b, c, d\}$. How many strings are there in Σ^4?
a) 256 b) 16
c) 4 d) 64

3) Which of the following is true while converting NFA to DFA?
I) Initial state of DFA will the initial state of NFA
II) Final state of DFA will be all sets of NFA's states that include at least one accepting state of NFA
a) Statement I) is false and statement II) is false
b) Statement I) is true and statement II) is false
c) Statement I) is false and statement II) is true
d) Statement I) is true and statement II is true

4) Given L = {ab. baa}*, which of the following is not in L?
a) ababaaab b) baaabbaa
c) abbaaab d) ababbaaab

5) In Moore machine, the length of output is _____
a) Same as length of input
b) Greater than length of input by one
c) Less than length of input
d) None of above

6) CFG S ⟶ AB | AS, A ⟶ a | aA, B ⟶ b generates the language _____
a) (ab)* b) a(ab)*b
c) aa*b⁺ d) aa*b

7) Which of the following grammars are in Chomsky Normal Form?
a) $S \longrightarrow AB \mid BC \mid CD, A \longrightarrow 0, B \longrightarrow 1, C \longrightarrow 2, D \longrightarrow 3$
b) $S \longrightarrow AB, S \longrightarrow BCA|0|l|2|3$
c) $S \longrightarrow ABa, A \longrightarrow aab, B \longrightarrow Ac$
d) All

8) Which of the following is acceptance of PDA using empty stack?
a) $(q_0, w, Z_0) \vdash^* (q, \varepsilon, \varepsilon)$
b) $(q_0, w, Z_0) \vdash^* (f, \varepsilon, \alpha)$
c) $(q_0, w, Z_0) \vdash^* (q, \varepsilon, \varepsilon)$ or $(q_0, w, Z_0) \vdash^* (f, \varepsilon, \alpha)$
d) None

9) In NPDA _____
a) There must be at most one choice of move for any state q, input symbol a, and stack symbol X
b) There must be zero or more choice of move for any state q, input symbol a, and stack symbol X.
c) There must be more than choice of move for any state q, input symbol a, and stack symbol X.
d) None

10) Which of the following is the Pumping Lemma conditions for $z = uvwxy$?
a) $|vwx| \leq n, |vx| > 0$, and for all $i \geq 0, uv^iwx^iy$ is in L
b) $|vwx| > n, |vx| > 0$, and for all $i \geq 0, uv^iwx^iy$ is in L
c) $|vwx| \equiv n, |vx| > 0$, and for all $i \geq 0, uv^iwx^iy$ is in L
d) None

11) Turing machine is more powerful than:
a) Finite automata
b) Push down automata
c) Finite automata and Push down automata
d) None of these

12) A Turing machine is expressed as a 7-tuple $(Q, \Sigma, \Gamma, \delta, q0, B, F)$ where Γ is complete set of tape symbols and Σ is a finite set of input symbols then what is always true?
a) $\Gamma \subseteq \Sigma$
b) $\Sigma \subseteq \Gamma$
c) $\Sigma = \Gamma$
d) None

13) What is the reason behind a Turing machine is more powerful than finite state machine FSM?
a) Turing machine head movement is continued to one direction.
b) Turing machine head moment is in both directions i.e. left moment and right moment as well.
c) Turing machine has capability remember arbitrary long sequence of input string.
d) All

14) Which of the following pairs have different expressive power?
a) DFA and NFA
b) Deterministic single tape Turing machine and Non-Deterministic single tape Turing machine
c) Deterministic PDA and Non-deterministic PDA
d) Single tap Turing machine and multi-tape Turing machine

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COMPUTER SCIENCE AND ENGINEERING
Theory of Computation (BTN03402)

Day & Date: Wednesday, 22-05-2024 Max. Marks: 56
Time: 10:00 AM To 01:00 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Section – I

Q.2 Attempt any Three **12**

a) Compare DFA, and NFA with example.

b) Give deterministic finite automata accepting following language
- i) Number of 1's is even and number of 0's is odd.
- ii) Number of 1's is odd and number of 0's is even.

c) Construct finite automata equivalent to the following regular sets
- i) 10 + (0 + 11)0* 1
- ii) (1 + 10 + 110)* 0

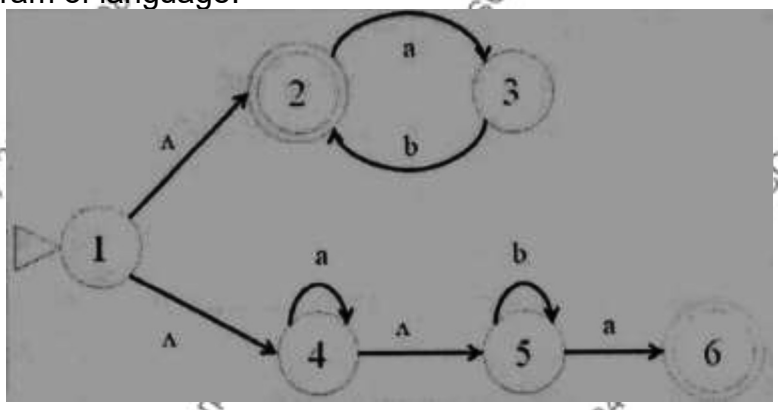
d) Construct a Moore machine that takes set of all strings over $\sum$ = {a, b} as input and prints 1 as output for every occurrence of baa as substring.

e) In each case what language is generated by a context free grammar.
- i) $S \longrightarrow aSa \mid bSb \mid \Lambda$
- ii) $S \longrightarrow aSa \mid bSb \mid a \mid b$
- iii) $S \longrightarrow aSb \mid bSa \mid \Lambda$
- iv) $S \longrightarrow aS \mid bS \mid a$

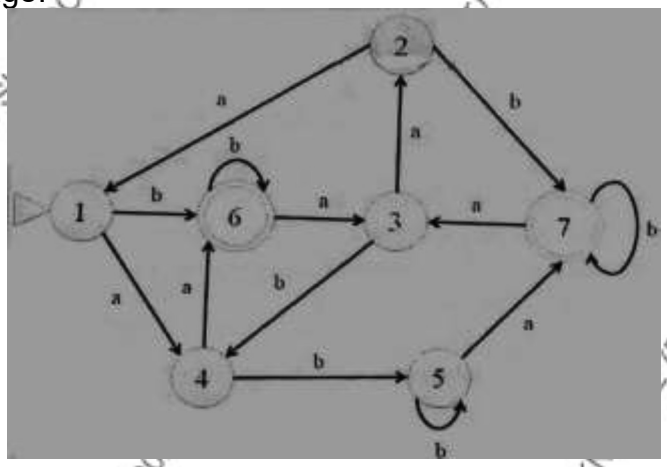
Q.3 a) Consider the language $L1$ where each string ends with 01 and the language $L2$ where each string ends with 10 over $\Sigma = \{0, 1\}$. Construct finite automata for $L1 \cup L2$, $L1 - L2$, $L2 - L1$ **08**

OR

b) For the following diagram draw deterministic finite automata accepting the same diagram of language.



Q.4 For following finite automata, find a minimum state finite automata recognized the same language. **08**



Section – II

Q.5 Attempt any Three. **12**

a) Define PDA with example.

b) State and explain the Pumping Lemma for CFL's.

c) Construct a Turing machine for L= $\{a^n b^n \mid n>0\}$.

d) Design TM for {a,b}*aba

Q.6 a) Prove that the language $L = \{a^n\, b^n\, c^n \mid n >= 0\}$ is not CFL using Pumping Lemma. **08**

OR

b) Design PDA for even and odd palindrome.

Q.7 Write short notes on following: **08**

a) Multi-tape Turing machine

b) Universal Turing machine

c) Non-deterministic Turing machine

d) Multi-track Turing machine

22/05/2024 08:47 AM SKNSCOE (Pandharpur)